

Day: 15

* What are Windows Event Logs?

- Windows records everything that happens on a system in Event Logs. For a DFIR analyst these logs are like a black box recorder — they tell you who logged in, what ran, what failed, and when.

* Event Log Locations

Log	Path	Contains
Security	Windows Logs\Security	Logons, privilege use, object access
System	Windows Logs\System	OS events, service start/stop
Application	Windows Logs\Application	App crashes, errors
PowerShell	Apps & services \ Microsoft\Windows\ PowerShell	PS command run
Sysmon	Apps & Services \ Microsoft\Windows\ Sysmon	Deep system monitoring

* The Most Important Event IDs

1) Authentication Events:

Event ID	Meaning	DFIR Relevance
4624	Successful logon	Who logged in and when
4625	Failed logon	Brute force attempts
4634	Logoff	Session Ended
4648	Logon with explicit credentials	Lateral movement
4672	Special privileges assigned	Admin logon
4768	Kerberos ticket requested	Domain auth
4771	Kerberos pre-auth failed	Failed domain logon

2) Account Management Events:

Event ID	Meaning	DFIR Relevance
4720	User account created	Attacker creating backdoor account
4722	User account enabled	Dormant account activated

4724	Password reset attempt	Privilege escalation
4728	User added to security group	Privilege escalation
4732	User added to local admin group	Critical- attacker gaining admin access

3> Process and Execution Events:

Event ID	Meaning	DFIR Relevance
4688	New process created	What ran on the machine
4689	Process exited	Process lifetime
1102	Audit log cleared	Attacker covering tracks!
4698	Scheduled task created	Persistence mechanism
7045	New service installed	Malware installing itself

* Logon Types - Critical for DFIR

- When we see Event ID 4624, the Logon Type tells us How

the user logged in:

Logon Type	Name	Meaning
2	Interactive	Local keyboard login
3	Network	File share, net use
4	Batch	Scheduled task
5	Service	Windows service
7	Unlock	Screen unlock
10	RemoteInteractive	RDP login
11	CachedInteractive	Offline cached login

* Note: Logon Type 10 = RDP — always investigate in an incident!

* Reading a 4624 Event

Event ID: 4624
 Time: 2024-04-14 03:42:17
 Account Name: Abhishek
 Account Domain: DESKTOP - XYZ
 Logon Type: 10 ← RDP — suspicious at 3am!
 Source IP: 185.220.X.X ← External
 IP = very suspicious!
 Workstation: ATTACKER - PC

- This single event tells you:

1) Someone RDP'd into this machine at 3:42 AM

2) From an external IP

3) Named their machine "ATTACKER-PC"

* Attack Timeline Using Event Logs

- A typical attack leaves this exact trail:

- 4625 x 50 → Brute force (50 failed logons)

- 4624 Type 3 → Network logon succeeded

- 4728 → Added to security group

- 4732 → Added to local admin group

- 4688 → Malicious process launched

- 4698 → Scheduled task created (persistence)

- 1102 → Logs cleared (covering tracks)

* PowerShell Event Logging

- 4103 → Pipeline execution

- 4104 → Script block logging → full

command visible

Note: Event 4104 is gold — it shows exactly what PowerShell command was run, even obfuscated ones.

* How to Access Event Logs

- Method 1 :: Event Viewer (GUI):

Win + R → eventvwr.msc

Windows Logs → Security → Filter Current Log → Event ID

- Method 2 : Powershell:

Get last 20 security events

Get-EventLog -LogName Security -Newest 20

Get specific Event ID

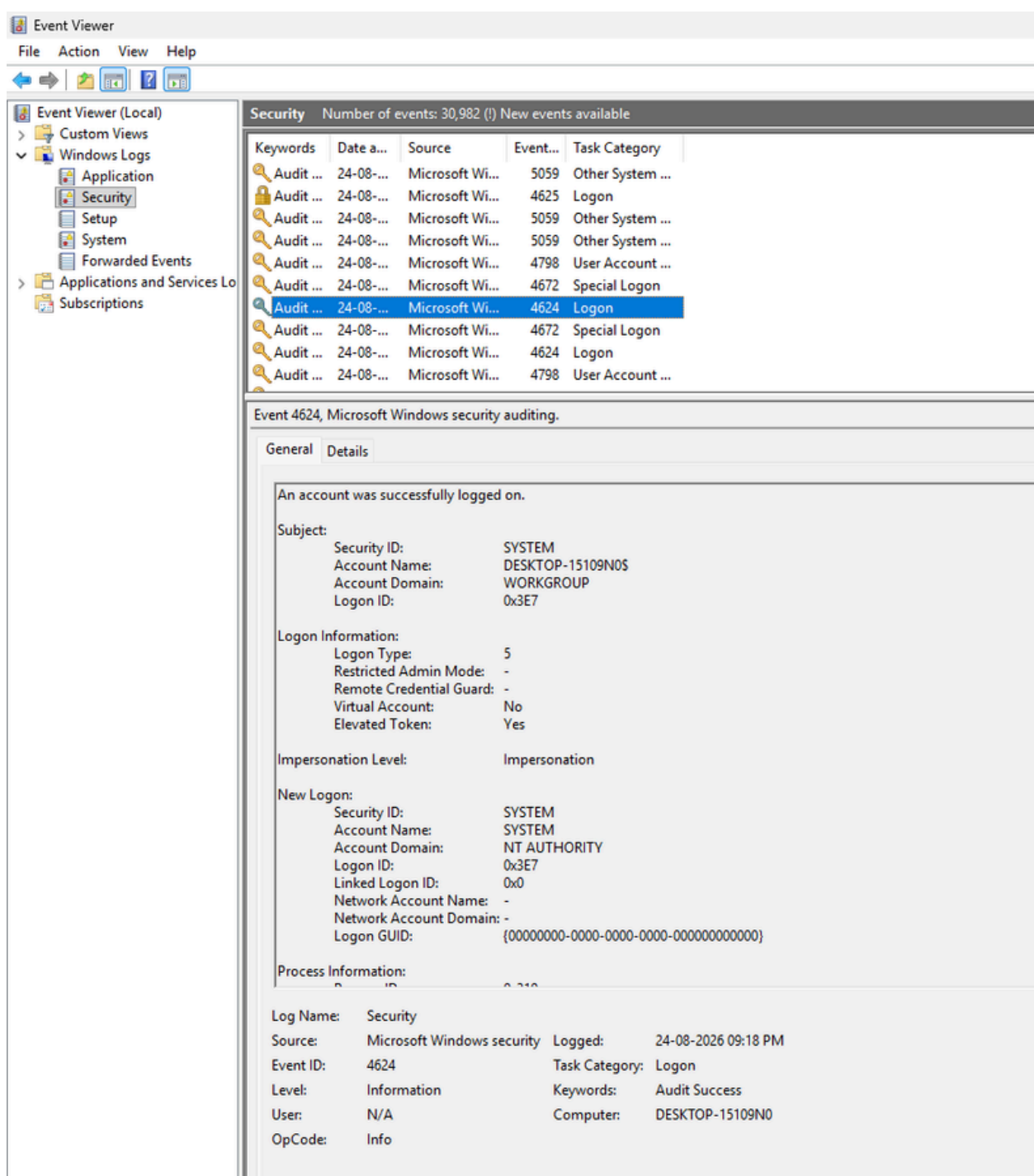
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4624} -MaxEvents

Get failed logons

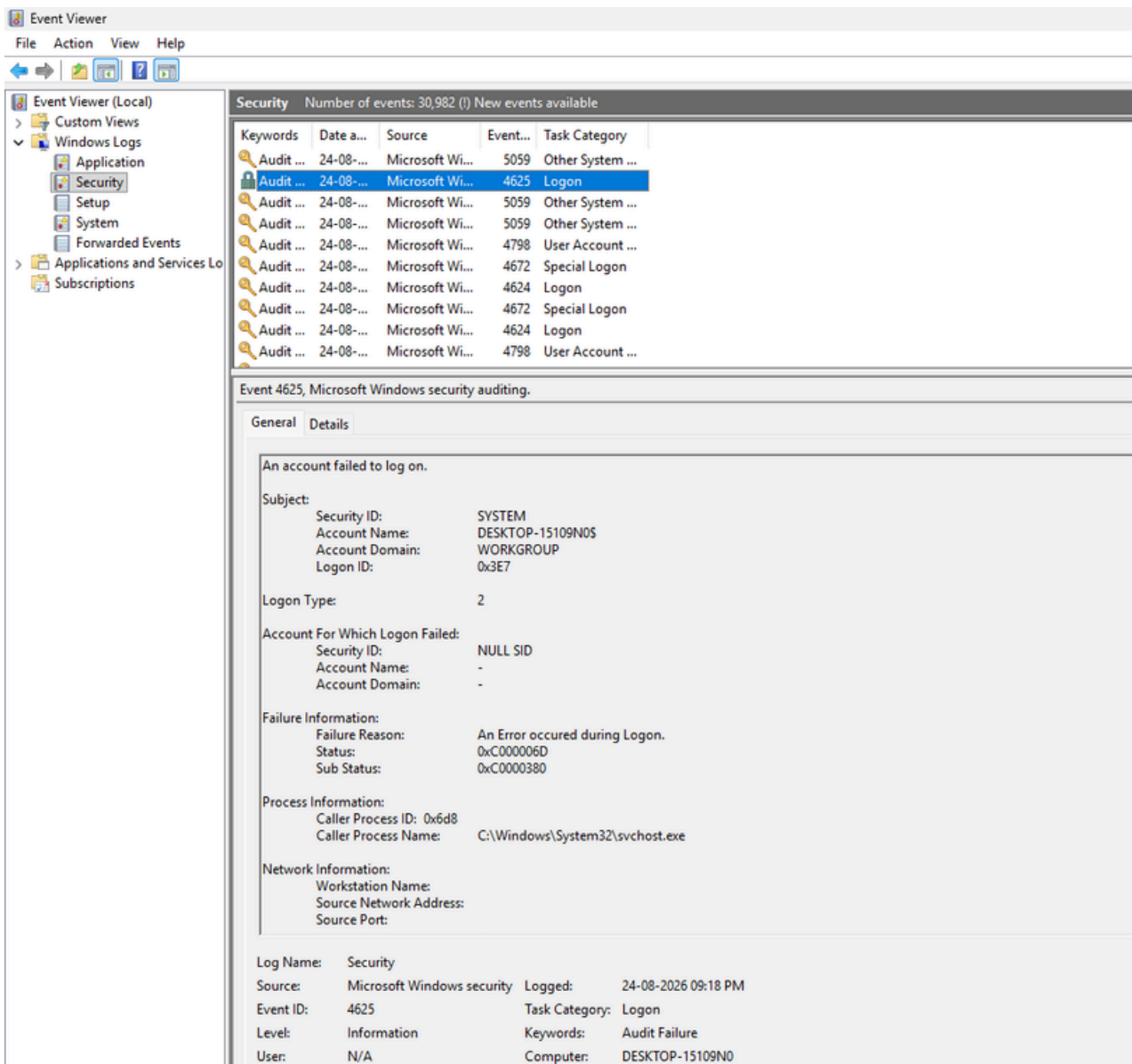
Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625} -MaxEvents

Part 1 – Event Viewer:

1. Press Windows + R → type eventvwr.msc → Enter
2. Navigate to: Windows Logs → Security
3. Find these Event IDs and click each one:
 - 4624 – Successful logon



- 4625 — Failed logon



Part 2 — PowerShell:

1. Open PowerShell as Administrator
2. Run:

Get-WinEvent -FilterHashtable

@{LogName='Security'; Id=4624} -MaxEvents 5

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4624} -MaxEvents 5

ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated          Id LevelDisplayName Message
-----
24-08-2026 09:27 PM 4624 Information An account was successfully logged on....
24-08-2026 09:21 PM 4624 Information An account was successfully logged on....
24-08-2026 09:18 PM 4624 Information An account was successfully logged on....
24-08-2026 09:18 PM 4624 Information An account was successfully logged on....
24-08-2026 09:18 PM 4624 Information An account was successfully logged on....
```

Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} -MaxEvents 5

```
PS C:\WINDOWS\system32> Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} -MaxEvents 5

ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated          Id LevelDisplayName Message
-----
24-08-2026 09:18 PM 4625 Information An account failed to log on....
24-08-2026 09:18 PM 4625 Information An account failed to log on....
24-08-2026 09:18 PM 4625 Information An account failed to log on....
23-08-2026 02:36 PM 4625 Information An account failed to log on....
20-08-2026 09:42 PM 4625 Information An account failed to log on....
```

Get-EventLog -LogName Security -Newest 20

```
PS C:\WINDOWS\system32> Get-EventLog -LogName Security -Newest 20

Index Time          EntryType Source                                InstanceID Message
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1266232 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266231 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266230 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266229 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266228 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266227 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266226 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266225 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266224 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266223 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266222 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266221 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266220 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266219 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266218 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266217 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266216 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266215 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266214 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
1266213 Aug 24 21:27 SuccessA... Microsoft-Windows... 5379 Credential Manager credentials were read....
```